Classification of Legal Text

Using NLP and Machine Learning

Course: UE24CS352A – Machine Learning

NAME: SAATHVIK SAI SUVARNA - PRITHVI RAJ

SRN: PES1UG24AM238 -PES1UG24AM204

1. Problem Statement

Legal opinions from the US Supreme Court are lengthy, complex documents written in non-standard English. Manually classifying these documents into legal categories is time-consuming and expensive.

This project aims to automatically classify US Supreme Court legal opinions into 14 predefined legal issue areas using NLP and Machine Learning techniques, specifically:

- **LDA + Logistic Regression** (baseline)
- **Doc2Vec + Logistic Regression** (improved)

2. Dataset Details

Source	US Supreme Court Database via textacy.datasets.SupremeCourt() (Washington University SCDB)
Size	8,398 legal opinions
Period	1946 – 2018 (Modern Decisions)
Type	Case-Centered, Legal Organization
Format	Full text of court opinions + metadata (issue_area, case_name, decision_date, etc.)
Train/Test Split	80% / 20% (6,717 / 1,680 cases)

Labels (14 categories)

Category	Count
Criminal Procedure	1,923
Economic Activity	1,667
Civil Rights	1,356
Judicial Power	1,135
First Amendment	658
Federalism	367
Unions	345
Due Process	335
Federal Taxation	304
Privacy	110

Category	Count
Attorneys	98
Interstate Relations	58
Miscellaneous	19
Private Action	1

3. Approach

STEP 1 Data Collection

- Legal opinions loaded via textacy SupremeCourt dataset

STEP 2 Preprocessing

- Lowercase conversion
- Special character removal
- Tokenization using spaCy
- Stopword removal
- Lemmatization
- Short word filtering (len > 2)

STEP 3 Model 1 – LDA + Logistic Regression

- Build TF-IDF dictionary (31,469 terms)
- Train LDA with 20 topics
- Convert docs to topic probability vectors
- Train Logistic Regression classifier

STEP 4 Model 2 – Doc2Vec + Logistic Regression

- Tag each document
- Train Doc2Vec (vector_size=100, epochs=40)
- Extract 100-dim document embeddings
- Train Logistic Regression classifier

STEP 5 Evaluation

- Compare accuracy, precision, recall, F1
- Confusion matrix analysis
- Per-class performance analysis

4. Implementation Overview

Libraries Used

Library	Purpose
textacy	Dataset loading
spaCy	NLP preprocessing
gensim	LDA and Doc2Vec models
scikit-learn	Logistic Regression + metrics
pandas	Data manipulation
matplotlib	Visualizations
seaborn	Confusion matrix plots

Platform	Google Colaboratory
Language	Python 3.13

5. Results

Overall Accuracy

Model	Accuracy
LDA + Logistic Regression	64.35%
Doc2Vec + Logistic Regression	68.99%

Doc2Vec + LR – Best Per-Class Results

Category	Precision	Recall	F1
Criminal Procedure	0.80	0.85	0.82
First Amendment	0.72	0.86	0.78
Unions	0.74	0.86	0.79
Federal Taxation	0.65	0.87	0.74
Civil Rights	0.69	0.69	0.69
Economic Activity	0.74	0.66	0.70

Key Observations

- Doc2Vec outperforms LDA by 4.64%
- High-volume classes (Criminal Procedure, Economic Activity) perform best
- Low-volume classes (Miscellaneous, Interstate Relations) perform worst
- LDA topics are interpretable and meaningful (e.g., Topic 15: trial, jury, defendant, counsel → Criminal Procedure)

6. Conclusions

1. NLP techniques successfully automate legal text classification with ~69% accuracy.
2. Doc2Vec embeddings capture semantic meaning better than LDA topic distributions.
3. Class imbalance significantly affects performance on minority categories (Privacy: 110 cases, Attorneys: 98 cases).
4. LDA topics are highly interpretable and align well with legal categories.
5. Future improvements:
    - Use BERT/Legal-BERT for better accuracy
    - Address class imbalance with SMOTE
    - Experiment with more LDA topics
    - Try ensemble methods